

Exercise 8 from the [chapter 12 exercises](#).

Waiting at an ER. A 95% confidence interval for the mean waiting time at an emergency room (ER) of (128 minutes, 147 minutes). Answer the following questions based on this interval.

A local newspaper claims that the average waiting time at this ER exceeds 3 hours. Is this claim supported by the confidence interval? Explain your reasoning.

No, this claim is not supported, because a 95% confidence interval estimates a range where the true population mean is most likely to fall. Since 3hrs or 180min is outside of our interval this claim is not supported.

The Dean of Medicine at this hospital claims the average wait time is 2.2 hours. Is this claim supported by the confidence interval? Explain your reasoning.

Yes, this claim is supported because 2.2hrs = 132 minutes which falls within our interval. Since the confidence interval contains this value, it is possible that the true mean could be 132minutes.

Without actually calculating the interval, determine if the claim of the Dean from part (b) would be supported based on a 99% confidence interval?

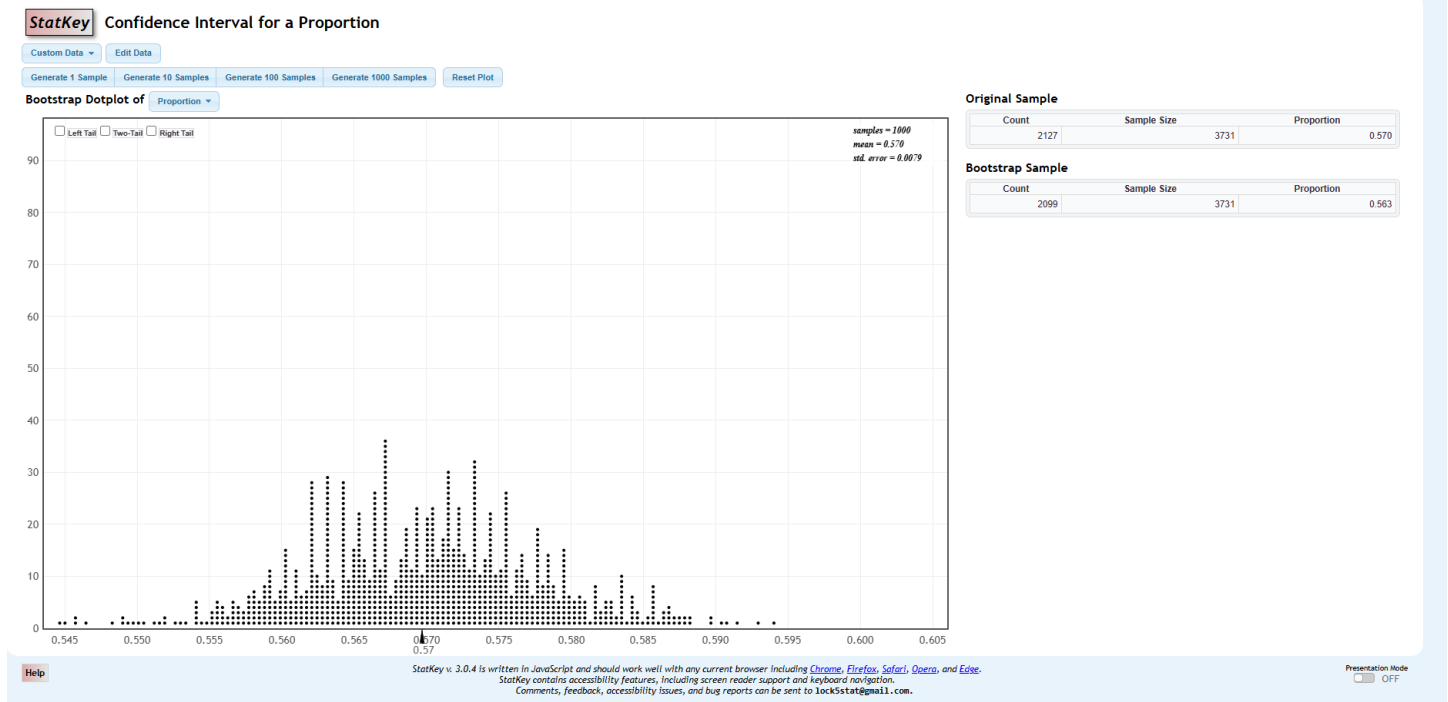
Yes, the 99% confidence interval would be wider than the 95% confidence interval. Since the 95% interval already has 132 min the 99% interval will likely still have 132 and more values on either side so to restate yes the deans claim would be supported by a 99% confidence interval.

A Gallup poll surveyed 3, 731 randomly sampled US residents in April 2021, asking how they felt about requiring proof of COVID-19 vaccination for travel by airplane. The poll found that 57% said they would favor it.

The 57% is a statistic because it describes the sample. What is the corresponding parameter of interest, in words?

The proportion of all U.S. residents who would favor requiring proof of COVID-19 Vaccination for air travel in April 2021.

Use [StatKey](#) to create a bootstrap distribution for this sample data. Copy and paste a picture of your bootstrap distribution here. Make sure to include the StatKey logo in your picture (and the buttons near it) so I can check to see if you have used the appropriate applet!



Use your bootstrap distribution and the standard error from the graph to compute the 50% and the 95% confidence intervals. Be sure to tell me which is which.

50% interval: (0.564, 0.575)

95% interval: (0.554, 0.586)

Write two sentences of interpretation – one for each confidence interval. “We are ____ confident that ...”

50% CI: We are 50% confident that the true proportion of U.S. residents who favor requiring proof of COVID-19 vaccination for air travel is between 56.4% and 57.5%.

95% CI: We are 95% confident that the true proportion of U.S. residents who favor requiring proof of COVID-19 vaccination for air travel is between 55.4% and 58.6%.

Use the formula calculate the standard error and make sure it’s close to the one in the graph above!

$$SE = \sqrt{0.57(1-0.57)/3731} = 0.0081$$

In the National Health and Nutrition Examination Survey (NHANES), subjects were whether they had smoked at least 100 cigarettes in their lives. The researchers then asked the 2328 people who answered “yes” (our sample) the age at which they started smoking.

Identify some features of this study. (I filled out part of this for you!)

Population of interest: US residents who have smoked at least 100 cigarettes

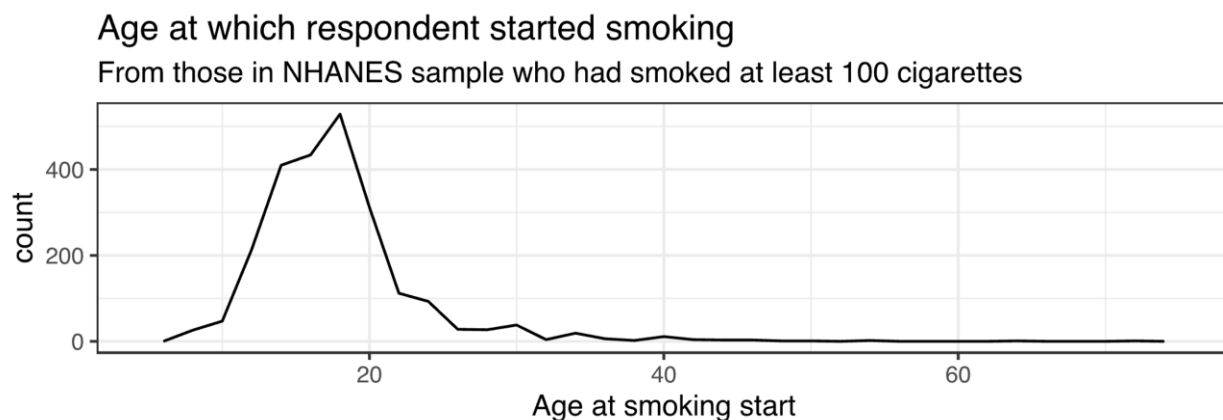
Variable of interest: the age of an individual that started smoking

Type of variable (categorical or numerical): numerical

What population parameter might we use the study data to estimate:

The mean (or median) age at which all US smokers started smoking.

We can summarize the distribution of ages at which people in the sample started smoking with our usual tools: a frequency polygon and some summary statistics.



0.025 quantile	Q1	median	Q3	0.975 quantile
11 years	15 years	18 years	20 years	31 years

mean	standard deviation
18.2 years	years

Describe the distribution of smoking-start ages in this sample of 2328 people who have smoked at least 100 cigarettes. As always, address the center, spread, and shape of the distribution. Include the context in your description.

The distribution of starting ages for people who have smoked at least 100 cigarettes is a right skew, with the majority of respondents starting between the ages of 15 and 20 years. The median is 18 years old, and the mean is 18.2 years showing its mostly symmetric with some late starters going to the upper end of 43 years old. The spread ranges from about 11 to 31 years for most individuals. And a standard deviation of 5.2 indicates moderate variability.

I have provided a CSV file containing the data from the study.

Load this data file into the StatKey bootstrap applet “CI for Single Mean, Median, St. Dev” > Upload File as we have done in class.

Use the applet to compute 50% and 95% bootstrap confidence intervals. Report them here.

50% CI: (18.128, 18.282)

95% CI: (17.972, 18.427)

Write sentences of interpretation.

I am 95% confident that the true mean age at which U.S smokers started smoking is between 17.972 and 18.427.

I am 50% confident that the true mean age at which U.S smokers started smoking is between 18.128 and 18.282.

Use the formula calculate the standard error and make sure it's close to the one in the graph above!

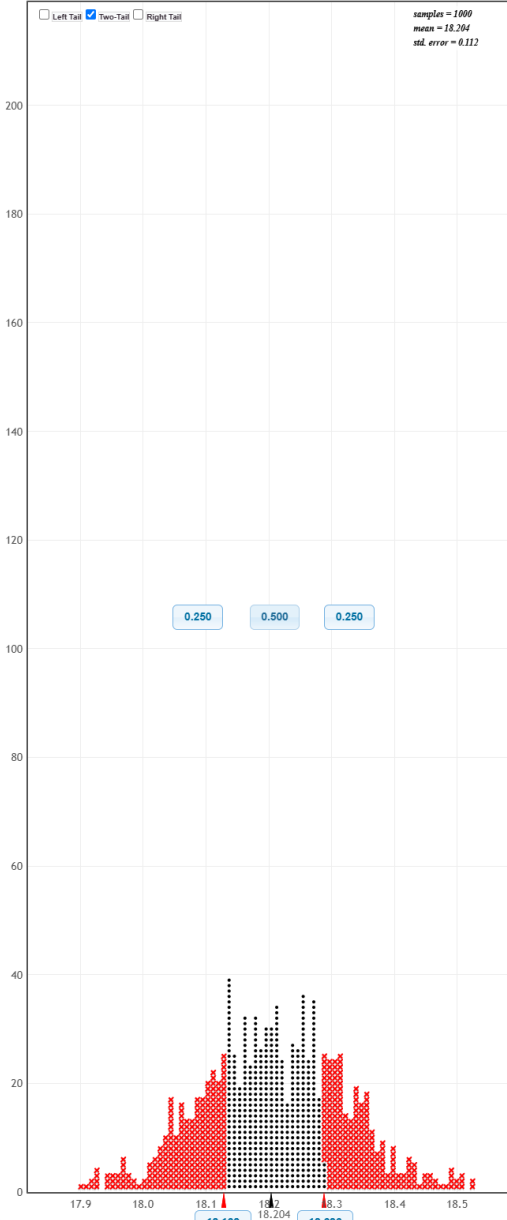
$$SE = 5.2/\sqrt{2328} = 0.108$$

- Pic if needed

StatKey Confidence Interval for a Mean, Median, Std. Dev.

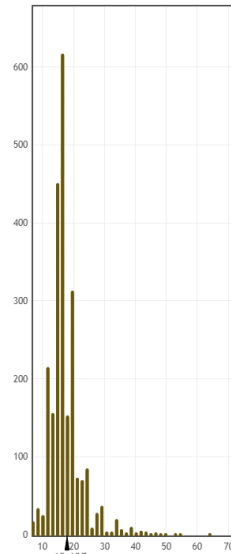
smoking_start.csv: age Show Data Table Edit Data Upload File Change Column(s)
Generate 1 Sample Generate 10 Samples Generate 100 Samples Generate 1000 Samples Reset Plot

Bootstrap Dotplot of



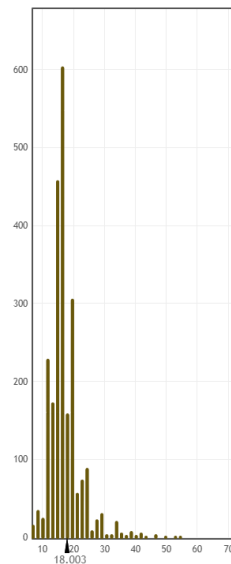
Original Sample

n = 2328, mean = 18.197
median = 18, stdev = 5.388



Bootstrap Sample Show Data Table

n = 2328, mean = 18.003
median = 17.5, stdev = 5.169



Help StatKey v. 3.0.4 is written in JavaScript and should work on any current browser including Chrome, Firefox, Safari, Opera, and Edge. StatKey contains accessibility features, including screen reader support and keyboard navigation. Comments, feedback, accessibility issues, and bug reports can be sent to lock5stat@gmail.com.

Presentation Mode
☐ OFF